

Digitally adjustable voltage source: D/A converter

A D/A converter (DAC, digital-to-analog converter): converts a digital signal sequence into an analog voltage.

One of the simplest implementations:
binary-weighted DAC (3 bits in the figure):

Binary numbers (b_0 , b_1 , b_2) are fed to the inputs. In practice the 0 or 1 binary input is represented by 0V or $V_S \sim 5V$ voltage at the input channels. The resistances of the channels increase according to the powers of 2.

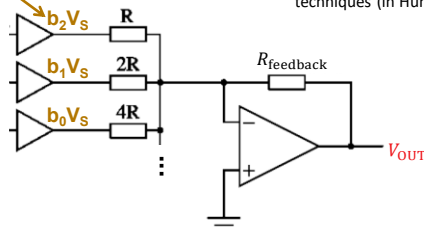
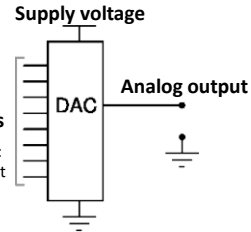


Figure sources: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)



Example with 3 bits:

The input is a number encoded in binary (e.g. 101). The summing circuit uses increasing resistors in powers of 2 to convert the binary input signal to an analog value.

Example: if the gains are adjusted to give an output of $V_{OUT}=1V$ for the least significant bit (001), then the output in volts is simply the decimal value of the input binary signal, e.g., an input of (101) gives an output of 5V. This device requires a single op-amp and N binary inputs for N bit resolution.

According to the operation of the summing amplifier the output voltage is:

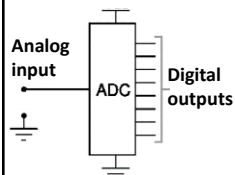
$$V_{OUT} = -\frac{V_S R_{feedback}}{2^N R} \sum_{i=0}^N 2^i b_i, \quad b_i = 0,1, V_S \approx 5V$$

i.e. the output is effectively a conversion of the input binary number to decimal number. The latter is scaled by the values of R, $R_{feedback}$ and V_S to the desired output voltage range.

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Digital voltmeter: A/D converter

A/D converter (ADC, analog-to-digital converter): converts an analog voltage into a digital (binary) signal. A small signal is first amplified and afterwards A/D conversion is applied.



Flash ADC: Equidistant voltage values are generated from a reference voltage V_{ref} using a resistor ladder, and these voltages are connected to the negative inputs of the A/D converter's comparator amplifiers (e.g. for $V_{ref}=7V$, the negative inputs are at 1,2, ..., 7 V, as shown in the figure below). Each comparator compares the voltage V_{in} applied to the input of the A/D converter with the reference voltage from the resistor ladder (e.g. for $V_{in}=5.2V$, the output of the bottom five amplifiers is V_{sat+} ("1" digital value) and the output of the other amplifiers is V_{sat-} ("0" digital value), i.e. a sequence of 111100 numbers is obtained, which is digitally converted to a binary number system, i.e. the number of "1" values (5 in this case) is converted to a binary number ($\rightarrow 101$).

The parallel (flash) ADC is perhaps the simplest in principle, the fastest, and also the most expensive (requiring the most components) converter. A 3-bit parallel ADC circuit is shown in the figure. Note, that for N bit resolution 2^N-1 comparators are required (7 comparators for 3bit resolution.)

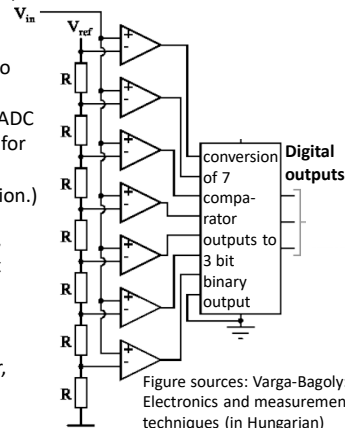


Figure sources: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

Successive approximation ADC: a single comparator amplifier is used, whose reference voltage is generated by a D/A converter. The output voltage of the DA converter is varied by a bisection method to find the digital voltage value closest to the input voltage based on the result of the comparator. (In the example first it is checked, if the input is larger or smaller than 4V (middle of the 0..8V range). If larger, the 4..8 V range is halved, i.e. the input is compared to 6 V. If smaller than 6V, the 4..6V range is halved, i.e. the input is compared to 5V.)

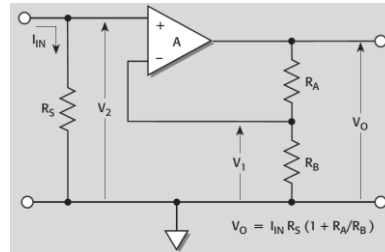
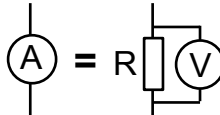
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Current measurement

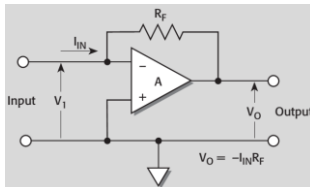
Source: Keithley Low Level Measurements Handbook

Shunt current meter circuit:

Simplest method: measure the voltage across a small resistor with a voltage amplifier.



Current amplifier circuit:



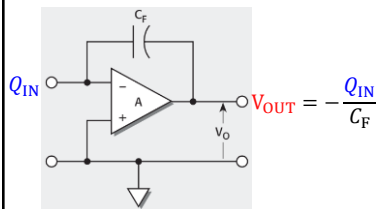
Similar to the inverting voltage amplifier circuit, just the R_2 resistor at the input is missing. The positive input is grounded, and therefore the negative input (i.e. the current input of the amplifier) is also at virtual ground (see rule (ii) of op-amps). According to rule (i) the entire input current flows towards the output, i.e. the output voltage is:

$$V_{OUT} = \frac{V_{IN}}{0} - I_{IN} \cdot R_F = -R_F \cdot I_{IN}$$

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Charge amplifier

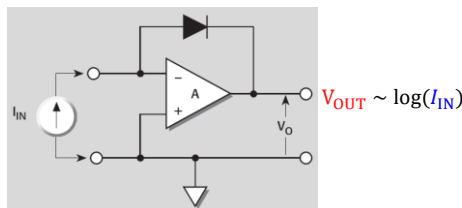
Source: Keithley Low Level Measurements Handbook



Again, the positive input is grounded, and therefore the negative input (i.e. the charge input of the amplifier) is also at virtual ground (see rule (ii) of op-amps). According to rule (i) no current can flow at DC operation, and the $Q=CV$ relation for a capacitor yields the demonstrated voltage at the output.

Note: in case of AC operation the capacitor can be considered as a circuit element with $Z=1/(i\omega C)$ complex impedance from which $V_{OUT} = -I_{IN}/(i\omega C_F)$ follows. This is practically a current amplifier with frequency dependent amplification and a 90-degree phase shift.

Logarithmic current amplifier



In the current amplifier circuit a diode is used instead of a feedback resistor, for which a $I \sim \exp(\text{const.} \cdot V)$ exponential current-voltage characteristic is observed. From this the output voltage is practically the logarithm of the input current. This amplifier is useful, if a very broad current range (many orders of magnitude) should be covered.

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Device specifications

Note the difference between accuracy and resolution: a ruler with a resolution of 1 mm does not necessarily have an accuracy of 1 mm.

In the figure the scale of the wooden ruler is misprinted, yielding 3mm error at 30cm reading. From this the accuracy is $1\text{mm} + 1\%$ of the reading.



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Analog inputs of a data acquisition card (DAQ)

Analog inputs (see 2AI lines)

ADC (Analog digital converter) resolution: 16 bit.

Voltagess in the range of $\pm 10\text{ V}$ are measured. This analog input voltage is converted to a digital signal, i.e. a 16 bit binary number. This means that the 20V input voltage range is divided to $2^{16} = 65536$ resolvable segments, i.e. the resolution for a 16 bit card and a $\pm 10\text{ V}$ input range is $20/65536\text{ V} = 305\mu\text{V}$

Possible input ranges: $\pm 10\text{ V}$, $\pm 2\text{ V}$ - in the $\pm 2\text{ V}$ input range first an 5x amplification is applied, and afterwards the ADC with $\pm 10\text{ V}$ input range digitizes the input signal.

Absolute accuracy – the error of the voltage measurements on the AI lines compared to the true values:

Nominal Range		Typical at 23 °C (mV)	Maximum (18 to 28 °C) (mV)
Positive Full Scale	Negative Full Scale		
10	-10	22.8	38.9
2	-2	4.9	8.6

Maximum sampling rate: 200kS/s –maximum 200000 voltage values are sampled in one second

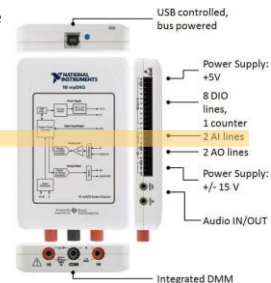
Input impedance $> 10\text{ G}\Omega$ – very high internal resistance

Overvoltage protection $\pm 16\text{ V}$ - at higher voltage the input is damaged.

The absolute maximum rating is the first to read in the device manual!

National Instruments myDAQ data acquisition module specifications

Analog Input:
2 channels, 200kS/s, 16-bit
Analog Output:
2 channels, 200kS/s, 16-bit
3.5mm stereo audio jacks
Digital I/O: 8 LVTTTL lines
Counter: 1 counter/timer
Integrated DMM: V, A, Ohm
Power Supply: $\pm 5\text{ V}$, $\pm 15\text{ V}$
Screw term + mass term option
Bus Powered (USB) operation



accuracy $\rightarrow$ **resolution** $\rightarrow$ **0.3mV**

Note the two orders of magnitude difference between the accuracy and the resolution! Still we can detect 0.3mV changes!

Common mode rejection ratio (CMRR) (DC to 60Hz): 70dB (see later)

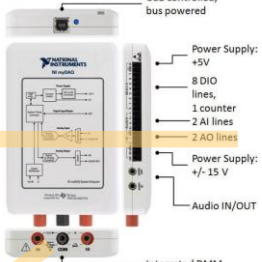
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Analog outputs of a data acquisition card (DAQ)

Analog outputs (see 2AO lines)
DAC (Digital to analog converter) **resolution: 16 bit**.
 Similarly, the $\pm 10\text{ V}$ output range is divided to $20\text{V}/2^{16} = 305\mu\text{V}$ steps.

Further specifications:
Range: $\pm 10\text{V}$, $\pm 2\text{V}$;
Maximum update rate: 200ks/s ;
Output impedance: 1Ω
Maximum output current: 2mA – e.g. a 1V output cannot supply $1\text{V}/10\Omega = 100\text{mA}$ to a 10Ω load, even if this load is larger than the output impedance. Instead, it can only output 2mA .

National Instruments myDAQ data acquisition module specifications



Analog Input:
2 channels, 200ks/s , 16-bit

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2 channels, 200ks/s , 16-bit
3.5mm stereo audio jacks

Digital I/O: 8 LVTTTL lines

Counter: 1 counter/timer

Integrated DMM: V, A, Ohm

Power Supply: $\pm 5\text{V}$, $\pm 15\text{V}$
Screw term + mass term option
Bus Powered (USB) operation

USB controlled, bus powered

Power Supply: $\pm 5\text{V}$
8 DIO lines, 1 counter, 2 AI lines, 2 AO lines

Power Supply: $\pm 15\text{V}$

Audio IN/OUT

Integrated DMM

Digital multimeter input (DC voltage measurement)

Resolution: 3.5 digit
 full digit: 0-9; half digit: 0-1,
 i.e. a 3 digit device displays in the 0-999-range, whereas a 3.5 digit in the 0-1999 range

Function	Range	Resolution	Accuracy
			$\pm ([\% \text{ of Reading}] + \text{Offset})$
DC Volts	200.0 mV	0.1 mV	$0.5\% + 0.2\text{ mV}$

3.5 digit: the 200mV range is divided to 2000 parts

If we measure 100mV in 200mV range: the accuracy is $100\text{mV} \cdot 0.5\% + 0.2\text{mV} = 0.7\text{mV}$ ($> 0.1\text{mV}$ resolution!!)

illustration of 3.5 digits



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Conversion of possible resolution specifications

PPM: part per million

Decibel (dB):

$20 \cdot \log_{10} \left(\frac{V_{res}}{V_{range}} \right)$

Example The V_{res} voltage resolution is 1% of the full voltage range (V_{range})

→

Percent	PPM	Digits	Bits	dB	Portion of 10V
10%	100000	1	3.3	-20	1 V
1%	10000	2	6.6	-40	100mV
0.1%	1000	3	10	-60	10mV
0.01%	100	4	13.3	-80	1mV
0.001%	10	5	16.6	-100	100 μV
0.0001%	1	6	19.9	-120	10 μV
0.00001%	0.1	7	23.3	-140	1 μV
0.000001%	0.01	8	26.6	-160	100 nV
0.0000001%	0.001	9	29.9	-180	10 nV

Further possible specification using the datasheet of a Keithley 2001 7.5 digit DMM

- Temperature coefficient: $\pm(\% \text{ of reading} + \% \text{ of range})/^{\circ}\text{C}$
- Accuracy on different time scales (this means that accuracy degrades over time, i.e. it is not a bad idea to recalibrate a sensitive instrument at certain intervals!).

DCV INPUT CHARACTERISTICS AND ACCURACY

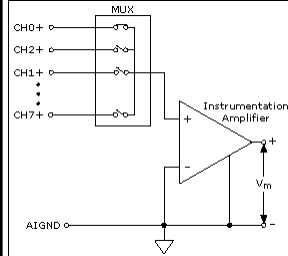
RANGE	FULL SCALE	RESOLUTION	DEFAULT RESOLUTION	INPUT RESISTANCE	ACCURACY ¹						TEMPERATURE COEFFICIENT $\pm(\text{ppm of reading} + \text{ppm of range})/^{\circ}\text{C}$ Outside $T_{CAL} \pm 5^{\circ}\text{C}$
					$\pm(\text{ppm of reading} + \text{ppm of range})$						
					5 Minutes ⁵	24 Hours ⁶	90 Days ⁷	1 Year ⁷	2 Years ⁷		
200mV ⁸	$\pm 210.0\text{mV}$	10nV	100nV	$>10\text{G}\Omega$	3 + 3	10 + 6	25 + 6	37 + 6	50 + 6	3.3 + 1.5	
2V	$\pm 2.10\text{V}$	100nV	1 μV	$>10\text{G}\Omega$	2 + 1.5	7 + 2	18 + 2	25 + 2	32 + 2	2.6 + 0.15	
20V	$\pm 21.0\text{V}$	1 μV	10 μV	$>10\text{G}\Omega$	2 + 1.5	7 + 4	18 + 4	24 + 4	32 + 4	2.6 + 0.7	
200V	$\pm 210.0\text{V}$	10 μV	100 μV	$10\text{M}\Omega \pm 1\%$	2 + 1.5	13 + 3	27 + 3	38 + 3	52 + 3	4.3 + 1	
1000V	$\pm 1100.0\text{V}$	100 μV	1mV	$10\text{M}\Omega \pm 1\%$	10 + 1.5	17 + 6	31 + 6	41 + 6	55 + 6	4.1 + 1	

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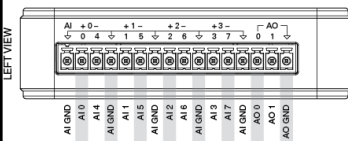
Possible DAQ card input types

It's important to note that most DAQ boards multiplex the inputs, i.e. one amplifier and one A/D converter measures all the inputs, with just switches to toggle which one is being measured. This also means that with a 200kS/s card, for example, a maximum sampling rate of 100kS/s per channel can be achieved by measuring 2 channels.

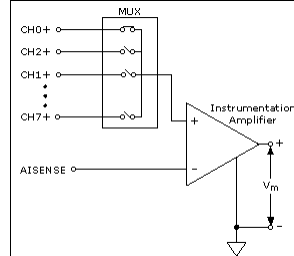
Referenced single ended input



Negative points for all channels are connected to AIGND, which is on the mains ground (e.g. we measure between AI0 and AIGND)

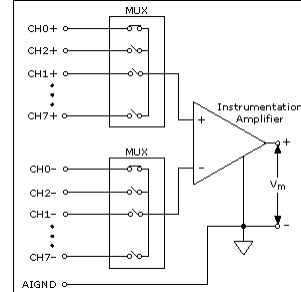


Non-referenced single ended input



Negative points for all channels are connected to AIGND, which is independent of the mains ground. (e.g. we measure between AI0 and AIGND, but in this setting the card internally disconnects AIGND with a switch from the mains ground)

Differential input:



True differential measurement, the card measures the voltage difference between two inputs (e.g. AI0 and AI4, or AI1 and AI5). This halves the number of input channels!

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Common mode rejection

Common mode rejection ratio (CMRR), e.g. for the analog input of a MyDAQ card: CMRR (DC to 60Hz): 70dB

Ideally, the output voltage of an operational amplifier depends only on the difference in input voltages. In the real case, however, if both inputs are around the same non-zero voltage, called common mode, a finite output voltage will appear. With a simple model:

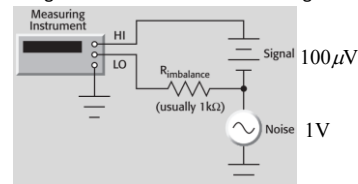
The suppression of the common mode signal is characterized as:

$$\text{CMRR} = \left| \frac{A_D}{A_{CM}} \right|; \quad \text{CMRR [dB]} = 20 \log_{10} \left(\left| \frac{A_D}{A_{CM}} \right| \right)$$

Conceptually, CMRR tells you how accurately small voltage differences can be measured far away from zero voltage, in the example figure the question is, how accurately a small (100μV) signal can be measured above a large background voltage (1V). CMRR may depend on frequency, so it is often given as a function of frequency.

$$V_{\text{out}} = A_D \cdot (V_+ - V_-) + \frac{1}{2} A_{CM} (V_+ + V_-)$$

Differential gain Common mode gain



Example: to measure a voltage difference of 100μV between HI and LO inputs with a DC background of around 1V, what is the measurement error at 70dB CMRR?

The instrument amplifies the measured signal before A/D conversion, according to the above model this amplified signal is: $V_{\text{out}} = A_D \cdot 100\mu\text{V} + A_{CM} \cdot 1\text{V}$. From this the measured signal is deduced as $V_{\text{measured}} = V_{\text{out}} / A_D$, i.e. the measurement error of the 100μV signal due to the common mode amplification is **ERROR = $1\text{V} \cdot A_{CM} / A_D = 1\text{V} / \text{CMRR}$**

$$\text{CMRR [dB]} = 70\text{dB} \rightarrow \text{CMRR} = 10^{3.5}$$

ERROR = $1\text{V} / 10^{3.5} = 316\mu\text{V}$, i.e. the error is larger than the signal to be measured. Without the 1V DC background signal, i.e. if the 100μV would be measured around zero voltage background, this error would not appear.

As a reference a **Keithley 2001 DMM with CMRR=140dB** would yield a much smaller error: **ERROR = $1\text{V} / 10^7 = 0.1\mu\text{V}$** !

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High speed measurements

So far the speed (or frequency) of the measurement was not an issue in the discussions. Next, we treat various aspects of high frequency measurements, such as:

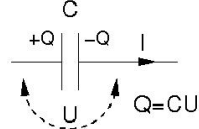
- Synchronization of signals, triggering strategies
- Special techniques for ultrafast sampling
- Aliasing
- RC time constant limitations
- Wave propagation in coaxial cables, transmission and reflection on circuit elements.



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Reminder: complex impedance

Capacitor:



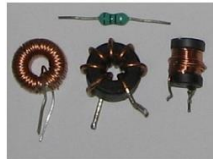
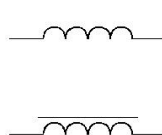
$$Q = CU$$

$$I = C \frac{dU}{dt}$$

$$U_{\omega}(t) = \hat{U}_0 e^{i\omega t} \Rightarrow I_{\omega}(t) = C \cdot i\omega \hat{U}_0 e^{i\omega t} = i\omega C \cdot U_{\omega}(t)$$

$$\Rightarrow U_{\omega}(t) = Z_C(\omega) I_{\omega}(t), \quad Z_C(\omega) = \frac{1}{i\omega C}$$

Inductor:

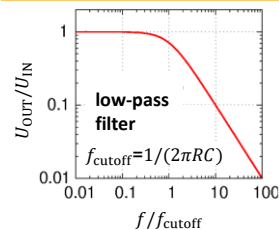
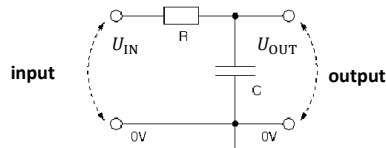


$$U = L \frac{dI}{dt}$$

$$I_{\omega}(t) = \hat{I}_0 e^{i\omega t} \Rightarrow U_{\omega}(t) = L \cdot i\omega \hat{I}_0 e^{i\omega t} = i\omega L \cdot I_{\omega}(t)$$

$$\Rightarrow U_{\omega}(t) = Z_L(\omega) I_{\omega}(t), \quad Z_L(\omega) = i\omega L$$

Example: RC filter



$$\frac{U_{OUT}}{U_{IN}} = \frac{Z_C}{Z_R + Z_C} = \frac{\frac{1}{i\omega C}}{R + \frac{1}{i\omega C}} = \frac{1}{1 + i\omega RC} \cdot \frac{1 - i\omega RC}{1 - i\omega RC} = \frac{1 - i\omega RC}{1 + (\omega RC)^2} \Rightarrow \left| \frac{U_{OUT}}{U_{IN}} \right| = \frac{\sqrt{1 + (\omega RC)^2}}{1 + (\omega RC)^2} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

Figure sources: Varga-Bagoly:
Electronics and measurement
techniques (in Hungarian)

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